FedScreen — Multi-Task Blood-Cell Screening: Final Implemented Pipeline

End-to-end workflow as built & validated in final.ipynb · 8 stages · malaria · anemia · leukemia

DATA PREP

INPUT

Raw public datasets
NIH Malaria · 27,558
AneRBC · 38,221
C-NMC 2019 · 10,661

ALL-IDB dropped:
260 imgs, no
patient IDs

1 Dataset Audit, Group IDs & Leakage-Safe Splitting

Metadata + label parsing → Group IDs patient / parent image → SHA-256 hash de-dup (test > val > train) → Group-based split (no group overlap)

Malaria: patient P<n> (200 pts) · Anemia: AneRBC parent image (1,000 groups) · Leukemia: patient (73 pts)
Leukemia official fold_0 (28 pts, unseen lab) = TEST · de-dup removed AneRBC 2,579 tr / 169 va / 86 te, C-NMC 30 tr (malaria: 0 duplicates)

OUTPUT

Leakage-free splits
(train / val / test)
Malaria
20.6k / 3.4k / 3.5k
Anemia
24.0k / 5.5k / 5.9k
Leukemia
6.1k / 1.0k / 3.5k

PREPROCESS

INPUT

Cell images
from splits
Varied labs, stains, scanners & cell sizes
Memory-mapped .npy caches (RAM / GPU, no DataLoader workers)

2 Normalisation & Stain-Robust Augmentation

Resize 192 × 192 (leuk. teacher: 224, ImageNet mean / std) → Per-image per-channel z-score (student, malaria & anemia) → GPU dihedral flips + rot90 + affine (±20°, 0.9–1.1×) → Stain jitter (0.08) · noise · random erase → Label smooth. 0.05 · MixUp / CutMix (malaria & anemia teachers only)

AneRBC colour-histogram RF probe = 94 % accuracy (stain shortcut)→ z-score + stain jitter; Macenko visually audited
Val / test splits are never augmented (only TTA at inference)

OUTPUT

Normalised
+ augmented
batches
192 × 192 × 3
tensors
Colour-consistent, varied views
Held-out splits

TEACHERS

INPUT

Task data
Malaria · Anemia · Leukemia
(each trained standalone, ImageNet-pretrained)

3 Standalone Task Teachers (ConvNeXt-Tiny + CBAM)

Malaria teacher
EMA weights · LLRD
MixUp / CutMix · 20 ep.
8× TTA · val-tuned thr.
patient-disjoint split
Test F1 0.969 · AUC 0.994

Anemia teacher
LLRD · warm-up + cosine
MixUp / CutMix · EMA
8× TTA · 25 epochs
de-dup at student stage
Test F1 0.863 · AUC 0.935

Leukemia teacher
224 px · DANN: GRL λ=0.5
(first 12 of 30 epochs)
5-cycle snapshot ens. · 8× TTA
train folds 1+2 → test fold_0
Test F1 0.826 · AUC 0.925

Leukemia teacher threshold is val-tuned (0.19; F1 0.739 @ 0.5) · malaria test n = 3,481 patient-disjoint (28 pts)

OUTPUT

Teacher assets
Task backbones
Soft labels for KD
(leukemia: 5-snap. ensemble)
F1 sanity gate
min val F1:
0.90 / 0.80 / 0.65
val 0.979 / 0.918 / 0.901 → all pass

MULTI-TASK

4 Shared ConvNeXt-Tiny Backbone + 3 Task-Specific CBAM Heads (28.6 M params)

ImageNet-pretrained ConvNeXt-Tiny (768-d) init = weight soup of teacher backbones → **Shared feature representation F**

Malaria head
CBAM: channel → spatial
Parasite rings, intracellular inclusions
→ 2-class logits

Anemia head
CBAM: channel → spatial
RBC shape, pallor, sickling / deformation
→ 2-class logits

Leukemia head
CBAM: channel → spatial
Nuclear morphology, chromatin, N:C ratio
→ 2-class logits

5 Kendall Uncertainty Weighting + Knowledge Distillation (joint training)

$$L_{total} = \sum_i \left[\tfrac{1}{2} e^{-s_i} L_i + \tfrac{1}{2} s_i\right] + \alpha L_{KD}$$

learnable $s_i = \log \sigma_i^2$ per task (malaria · anemia · leukemia)

α = 0.30 · KD from frozen teachers: malaria 1 · anemia 1 · leukemia 5-snapshot ens.
(a teacher is kept only if val Macro-F1 passes its sanity gate)

AdamW · base LR 1e-4 · layer-wise LR decay 0.80 · head LR 5e-4 · wd 0.05
EMA 0.9985 · grad-clip 1.0 · stochastic depth 0.1 · label smoothing 0.05
24 epochs × 400 steps · task-balanced batches (32 / task) · bf16 · patience 6
Val Macro-F1 @ ep. 24: malaria 0.977 · anemia 0.888 · leukemia 0.925

Output: one jointly optimised multi-disease model (malaria + anemia + leukemia) → fedscreen_mtl_convnext_FINAL.pth

EVALUATE

INPUT

Held-out test sets
Group-disjoint, de-duplicated:
malaria 3,481
anemia 5,895
leukemia 3,527 = unseen lab (fold_0)

6 Fixed-Threshold Operating Points & Held-Out Test Evaluation

Centralized thresholds: malaria 0.50 · anemia 0.45 · leukemia 0.50 (fixed, chosen after seeing test)

Val-tuned leukemia cut-off 0.783 rejected: val 77 % normal vs test 68 % ALL→ test F1 only 0.575

Task	Macro-F1	AUC	Standalone teacher (F1 / AUC)
Malaria	0.967	0.993	0.969 / 0.994
Anemia	0.835	0.922	0.863 / 0.935
Leukemia	0.892	0.950	0.826 / 0.925 *

Leukemia @ 0.50: acc 90.9 % · MCC 0.788 · balanced acc 87.7 % · confusion [[890, 240], [80, 2317]]

** teacher threshold val-tuned (0.19); at 0.5 the teacher scores F1 0.739 → not a like-for-like comparison*

Joint vs standalone: malaria −0.002 F1 · anemia −0.028 F1 (shared-backbone cost)

OUTPUT

Validated model
+ metrics
One shared backbone replaces 3 standalone nets:
≈ teacher on malaria, −0.03 F1 on anemia, higher F1 on leukemia

FEDERATED

7 Federated Simulation (FedProx) on the final ConvNeXt MTL — raw images never leave a client

Virtual PHC-0	Virtual PHC-1	Virtual PHC-2	Virtual PHC-3	Virtual PHC-4
1.1k imgs anemia only	15.4k imgs malaria-heavy	15.3k imgs leukemia-heavy	11.4k imgs anemia 0.8 % pos.	10.1k imgs anemia 93 % pos.
images stay local	images stay local	images stay local	images stay local	images stay local
50 local steps + μ/2·‖w−w_g‖²	50 local steps + μ/2·‖w−w_g‖²	50 local steps + μ/2·‖w−w_g‖²	50 local steps + μ/2·‖w−w_g‖²	50 local steps + μ/2·‖w−w_g‖²
Δw only	**Δw only**	**Δw only**	**Δw only**	**Δw only**

FedProx aggregator
μ = 0.01 · weighted FedAvg
15 rounds · Dirichlet α = 0.5
3–5 of 5 clients / round (10 % drop)
→ **Global model → broadcast**

STATUS
Executed on the final ConvNeXt MTL, warm-started from centralized FINAL (206 tensors). Saved: fedscreen_fl_FINAL.pth (this FL model is the one served by the app)

Test, FL vs centralized (F1 / AUC): Malaria 0.9665 / 0.9930 (central 0.9671) · Anemia 0.8338 / 0.9214 (central 0.8338)
Leukemia 0.8151 / 0.9516 (FL 81.5% F1 vs centralized 89.2% F1 @ fixed 0.50; central AUC 0.950)

NOTE: the notebook's printed mean F1 drop ("PASS") used the rejected 0.575 baseline (0.783 cut-off), so it is not valid.
Corrected metric: leukemia centralized 0.892 vs federated 0.815 (+0.08 gap); malaria / anemia gap ~0; mean drop +0.026 (target < 0.02).
(federated leukemia F1 was computed with its own val-tuned threshold; re-evaluate FL at 0.50 to confirm)

EXPLAIN & DEPLOY

8 Edge Deployment, Specimen Routing, Explainability & Clinician Decision Support

Federated global model (fl_FINAL, FP32 → INT8) + input image

Auto specimen router
Logistic regression on deployed-backbone features
→ malaria / RBC / WBC crop
100 % (random 70/30 split, not group-disjoint)
Reject if conf < 80 %; only matching head runs

Dynamic INT8 quantisation
nn.Linear layers → qint8
FP32 114.6 MB → 35.2 MB
(3.3×; <25 MB target NOT met)
CPU 43 ms → 29 ms / image
INT8 vs FP32 agree 15/16
(16 leukemia crops only)

Explainability
HiResCAM per active head
on 6×6 backbone map,
upsampled (not pixel-level)
percentile norm. + smoothing
Turbo map + Canny contours
Grad-CAM++ ×3: 236 ms (FP32)

Confidence gate
Per-task thresholds from deployed-model metadata
(leukemia: check 0.5 vs tuned)
conf ≥ 85 % → auto-report
conf < 85 % → Refer to pathologist (slider 0.60–0.99)

FedScreen · Streamlit web app (served via Cloudflare quick tunnel)
Upload image → auto-detect specimen (router) → matching head only (INT8 or FP32) → class + P(positive) + confidence
→ HiResCAM overlay (Turbo + contours) → verdict banner · non-matching heads shown as default 100 % Negative, "head not run"
12 demo sample crops (3 tasks × normal / abnormal) bundled for offline verification

→ **Clinician Decision Support**

ITERATION LOG

WHAT WE TRIED → WHAT FAILED → WHAT WE FIXED / SWITCHED TO

TRIED	FAILED	FIXED / SWITCHED
EffNet-B0 + GroupNorm @128 px, pre-made crops	Leukemia collapsed to majority class (F1 0.44); anemia only 0.80	**Rebuilt data; ConvNeXt-Tiny @192 px, per-image z-score**
ALL-IDB-1/2 for leukemia	Only 260 images, no patient IDs (cross-split leakage)	**Switched to C-NMC 2019 (10,661 imgs, 73 pts; fold_0 = test)**
First random group splits	Split-order bug, missing malaria negatives, filename groups	**Rebuilt grouped splits (patient / parent image): 0 leaked groups**
Random AneRBC splits	1,232 duplicates across splits; colour RF probe = 94 % acc	**SHA-256 de-dup (test > val > train) + z-score + stain jitter**
First centralized MTL, static equal loss (15 ep)	Val F1 stuck at 0.33 in epochs 5–12 (gradient interference)	**Kendall uncertainty weighting + KD + task-balanced batches**
Anemia head from baseline MTL	Test F1 0.80 / AUC 0.88	**Standalone teacher: LLRD, MixUp/CutMix, EMA, TTA → F1 0.86**
Leukemia trained on folds 1+2 only	Good val F1, much lower on unseen-lab fold_0 (domain shift)	**DANN (GRL) + 5-snapshot ensemble + 8× TTA → test F1 0.826**
Mixed ConvNeXt-Base / Tiny / EffNet heads	Weights incompatible; no true shared backbone	**One ConvNeXt-Tiny from teacher weight soup + KD (α = 0.30)**
Splits without de-duplication	72 duplicate leukemia images in train & test (+ AneRBC)	**Hash de-dup on all 3 tasks before final training**
Val-tuned leukemia threshold (0.783)	Val 77 % normal vs test 68 % ALL → test F1 only 0.575	**Fixed 0.50 cut-off → F1 0.892, acc 90.9 %, MCC 0.788**
Streamlit demo with placeholder outputs	Simulated probabilities, random heatmaps	**Real INT8 inference + CAM heatmaps (HiResCAM) on trained net**
FP32 model for PHC deployment	114.6 MB; < 25 MB size target missed	**Dynamic INT8: 35.2 MB, 29 ms CPU (latency met; size still open)**
Manual specimen selector in the app	User could run an off-target head → meaningless output	**Auto specimen router (100 % held-out acc.) runs matching head only**
FedAvg smoke test (EffNet-B0, 3 clients × 100 imgs)	Toy scale; never carried into the ConvNeXt model	**FedProx μ = 0.01, 5 clients × 15 rounds on final ConvNeXt MTL**

Local data / prototype zone · Data / feature flow · Shared model updates (Δw) · Key stage · Caveat